

# Microservice V/S monolithic

## 1 What is Monolithic Architecture?

A **monolithic architecture** is where:

The entire application is built and deployed as a single unit.

Everything runs in one codebase and one deployment.

### Structure Example

Client → Backend (Users + Orders + Payments + Auth) → Database

All modules share:

- Same runtime
- Same database
- Same deployment

### Example: Traditional SaaS App

- One Node.js project
- One database
- One server
- Single deployment pipeline

## ✓ Advantages of Monolith

- Simple to build
- Easy to debug
- Easy to deploy
- Less DevOps overhead
- Good for startups

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## ✗ Disadvantages

- Hard to scale specific modules
- Large codebase over time

- Slower deployments
- One bug can affect whole system

## **2 What is Microservices Architecture?**

Microservices break application into:

Small, independent services that communicate over network.

Each service:

- Has its own codebase
- Can have its own database
- Can be deployed independently

### **Structure Example**

Client → API Gateway → User Service  
→ Order Service  
→ Payment Service

Each service runs independently.

### **When to Use Monolith?**

Use monolith when:

- Startup stage
- Small team
- MVP
- Simple domain
- Limited traffic

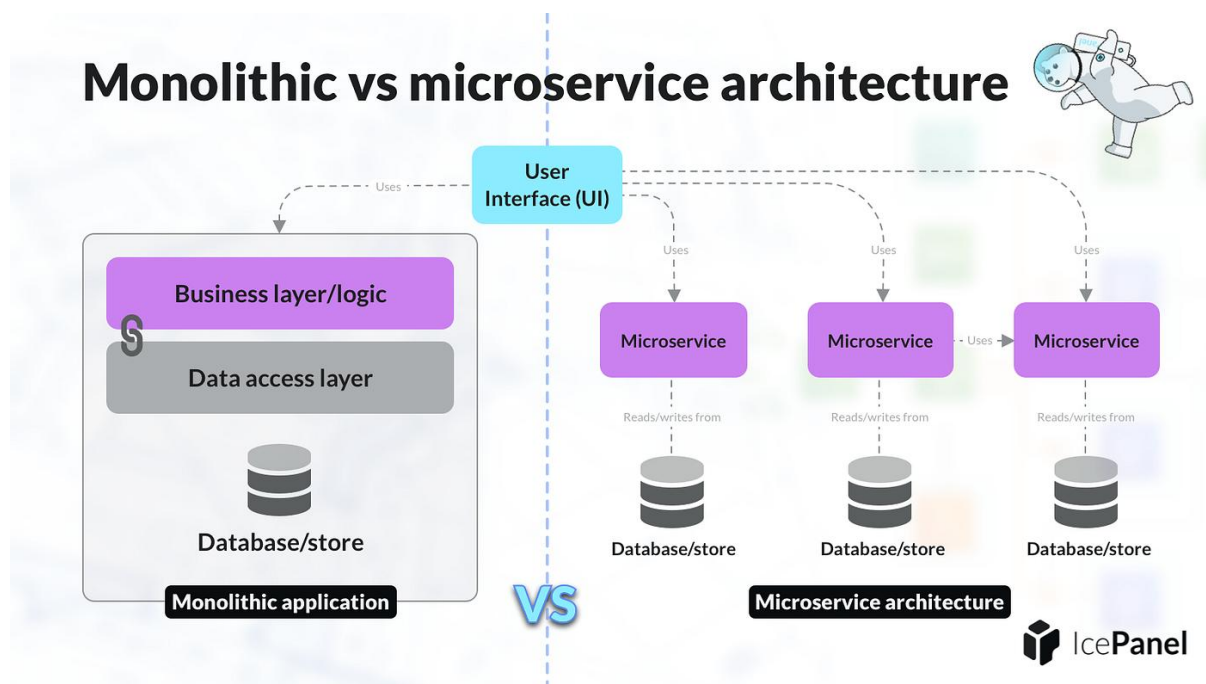
👉 Most successful startups started as monoliths.

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## 5 When to Use Microservices?

Use microservices when:

- Large engineering teams
- High traffic
- Complex domains
- Need independent scaling
- Frequent deployments



| Feature    | Monolith    | Microservices    |
|------------|-------------|------------------|
| Codebase   | Single      | Multiple         |
| Deployment | Single unit | Independent      |
| Scaling    | Entire app  | Service-specific |

| Feature         | Monolith       | Microservices    |
|-----------------|----------------|------------------|
| Database        | Usually shared | Usually separate |
| Complexity      | Low initially  | High             |
| DevOps          | Simple         | Advanced         |
| Fault isolation | Low            | Better           |

Monolithic architecture is a single unified codebase deployed as one unit. It's simpler to build and ideal for startups. Microservices architecture breaks the system into independent services that can scale and deploy separately. While microservices offer better scalability and fault isolation, they introduce distributed system complexity. For a growing SaaS product, I'd start with a modular monolith and migrate to microservices only when scaling demands it.